



PREFACE

The *MC68030 User's Manual* describes the capabilities, operation, and programming of the MC68030 32-bit second-generation enhanced microprocessor. The manual consists of the following sections and appendix. For detailed information on the MC68030 instruction set refer to M68000PM/AD, *M68000 Family Programmer's Reference Manual*.

- Section 1. Introduction
- Section 2. Data Organization and Addressing Capabilities
- Section 3. Instruction Set Summary
- Section 4. Processing States
- Section 5. Signal Description
- Section 6. On-Chip Cache Memories
- Section 7. Bus Operation
- Section 8. Exception Processing
- Section 9. Memory Management Unit
- Section 10. Coprocessor Interface Description
- Section 11. Instruction Execution Timing
- Section 12. Applications Information
- Section 13. Electrical Characteristics
- Section 14. Ordering Information and Mechanical Data
- Appendix A. M68000 Family Summary
- Index

NOTE

In this manual, assertion and negation are used to specify forcing a signal to a particular state. In particular, assertion and assert refer to a signal that is active or true; negation and negate indicate a signal that is inactive or false. These terms are used independently of the voltage level (high or low) that they represent.

The audience of this manual includes systems designers, systems programmers, and applications programmers. Systems designers need some knowledge of all sections, with particular emphasis on Sections 1, 5, 6, 7, 13, 14, and Appendix A. Designers who implement a coprocessor for their system also need a thorough knowledge of Section 10. Systems programmers should



become familiar with Sections 1, 2, 3, 4, 6, 8, 9, 11, and Appendix A. Applications programmers can find most of the information they need in Sections 1, 2, 3, 4, 9, 11, 12, and Appendix A.

From a different viewpoint, the audience for this book consists of users of other M68000 Family members and those who are not familiar with these microprocessors. Users of the other family members can find references to similarities to and differences from the other Motorola microprocessors throughout the manual. However, Section 1 and Appendix A specifically identify the MC68030 within the rest of the family and contrast its differences.